Ejercicio 2 - Procesamiento de Imágenes y Operador Sobel.

Este ejercicio realiza el procesamiento de una imagen de 96x96 píxeles aplicando operadores Sobel directamente en el MCU.

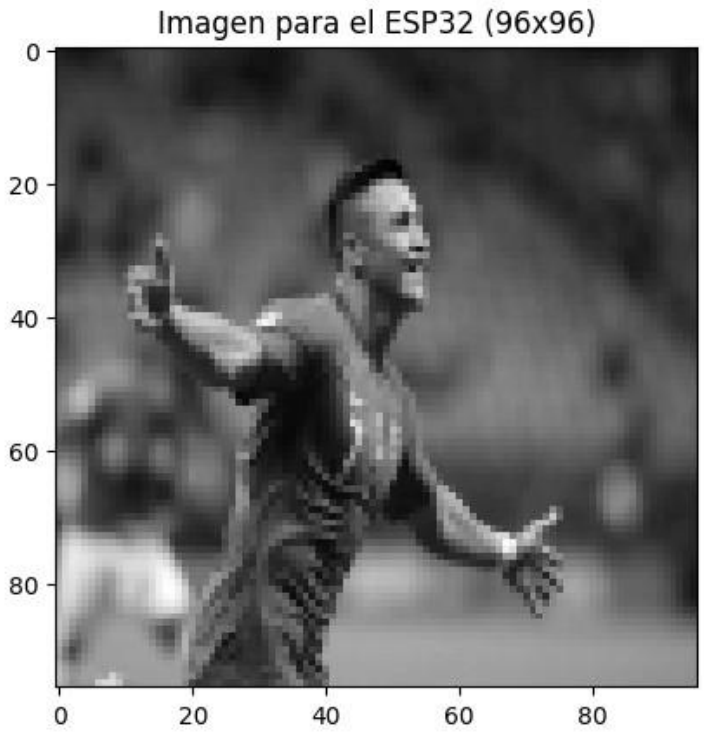
Para correr el código es necesario hacer los siguientes pasos:

PowerShell
usbipd list # Busca el BUSID de tu dispositivo
usbipd bind --busid <BUSID>
usbipd attach --wsl --busid <BUSID>

Terminal wsl
source ~/.profile && source ~/.bashrc && get_esp32 && idf.py
set-target esp32s3 && idf.py build && idf.py -p /dev/ttyUSB0
flash monitor

Link del google collab:
https://colab.research.google.com/drive/1elE3-fPI_UPIm27OYJbP31M0-mzsSUKO?usp=sharing
Foto original:

Foto monocromática y reescalada a 96x96



Resultados del ESP32 (Salida Serial):

diazb@DESKTOP-HGLIQB2: ~/Proyecto-Sumo-grupo-Juan-Lucas-Cristobal/Lab_1/Ejercicio_2

```
I (270) main_task: Calling app_main()
--- INICIANDO PROCESAMIENTO SOBEL (ALEXIS) ---
I (5270) SOBEL_PROJECT: Iniciando procesamiento Sobel...
--- INICIO DATOS SOBEL ---
0,0,0,
```

diazb@DESKTOP-HGLIQB2: ~/Proyecto-Sumo-grupo-Juan-Lucas-Cristobal/Lab_1/Ejercicio_2

```
4,52,68,64,60,64,58,52,46,48,56,48,46,64,78,54,38,90,102,56,46,70,78,74,60,50,56,52,0,
0,158,198,228,226,78,140,242,174,206,244,214,106,188,236,156,90,74,36,148,118,50,58,68,120,124,130,208,434,336,66,48,36,56,44,34,6,152,136,162,178,4,296,326,290,186,198,332,282,154,334,598,184,444,296,104,198,52,64,58,64,64,64,64,74,78,6
5,62,84,76,60,62,62,56,50,50,52,58,56,58,64,28,52,94,140,176,132,40,46,64,64,64,56,58,86,60,0,
0,122,190,244,268,136,96,206,192,226,262,210,126,204,260,222,142,84,84,136,90,30,32,50,104,144,68,40,266,240,98,54,48,62,66,38,40,46,210,76,216,136,260,216,294,222,136,158,180,94,358,466,100,378,452,128,246,162,36,50,58,54,62,58,60,70,72
,60,62,68,44,58,54,50,44,48,54,68,72,70,54,50,122,112,112,178,186,94,12,56,66,64,76,82,70,0,
0,60,144,234,286,166,62,106,128,180,200,136,94,150,204,214,194,172,146,154,86,26,18,50,90,120,102,80,274,188,86,54,104,114,116,82,62,36,34,180,38,122,120,164,86,190,126,108,148,124,446,334,34,218,450,388,70,268,74,24,30,22,34,34,40,40,48
,40,44,46,36,46,34,30,36,38,48,64,68,56,70,120,140,100,76,150,210,150,36,40,56,74,84,88,74,0,
0,14,96,214,284,168,54,36,64,106,108,56,42,74,98,124,168,208,186,178,132,66,54,88,108,134,130,130,364,232,54,122,136,114,88,44,100,50,68,42,160,206,48,114,170,190,44,126,196,252,444,178,4,68,310,498,208,288,400,106,38,10,12,18,24,20,28,2
0,24,16,24,32,16,18,22,22,30,42,44,20,84,174,158,110,82,124,222,206,70,42,58,64,68,70,56,0,
5,34,124,266,294,146,72,46,42,56,46,26,36,54,18,24,100,192,198,188,150,114,86,102,112,132,186,210,418,296,28,116,26,134,88,38,78,138,120,192,110,194,212,108,84,124,30,48,200,308,296,40,8,12,140,410,490,312,502,568,276,36,24,26,22,22,28,2
2,12,12,40,26,10,10,10,14,24,30,22,20,124,198,166,110,72,92,208,240,114,44,44,44,42,34,24,0,
0,86,178,300,268,114,68,64,56,54,54,44,56,74,48,48,98,112,156,150,148,128,96,94,110,114,236,218,382,368,100,10,116,60,42,52,116,16,110,152,382,202,90,170,108,232,122,84,138,296,168,10,2,2,34,234,570,434,76,508,610,258,32,26,22,28,34,34,2
6,28,34,4,10,6,12,30,40,32,16,22,118,186,154,88,34,56,158,222,136,48,38,38,24,6,4,0,
0,138,202,282,214,90,52,54,56,52,54,52,60,98,114,136,132,100,70,98,128,138,112,92,112,50,258,196,290,318,100,76,44,50,88,34,92,108,172,202,90,326,240,210,204,114,56,94,128,192,78,14,2,26,62,142,460,480,188,46,508,584,198,14,14,20,24,28,3
4,40,22,6,4,124,144,182,56,36,28,46,136,212,164,58,24,102,190,222,150,62,42,38,18,14,4,0,
0,170,174,220,152,80,54,44,48,60,84,102,102,158,192,176,138,100,20,46,94,134,128,108,102,94,266,134,274,174,106,66,68,64,120,118,48,72,30,120,206,116,50,98,180,196,222,26,60,114,32,10,46,88,96,88,386,444,120,160,64,536,508,134,8,6,10,18,
36,50,26,10,96,440,324,462,130,28,32,50,128,234,228,120,104,164,204,222,166,72,42,36,16,20,12,0,
0,162,92,144,118,84,64,54,64,128,160,168,198,238,228,148,140,134,82,72,66,86,104,98,66,170,236,70,222,184,138,28,104,88,152,212,112,78,108,172,148,70,22,92,128,30,244,114,74,192,124,116,122,94,40,16,352,508,78,104,200,96,544,470,160,16,2
0,28,50,66,154,344,524,572,216,456,152,12,12,32,88,210,262,180,156,178,186,200,150,56,26,18,8,20,8,0,
0,252,182,282,250,182,126,92,120,162,184,200,216,214,168,88,116,118,112,98,60,42,66,72,62,210,176,108,178,166,66,96,132,178,176,286,298,138,62,32,56,88,24,86,100,14,122,94,62,152,152,172,150,86,24,20,274,544,200,162,14,144,160,554,568,31
6,114,38,60,192,476,582,292,158,500,484,90,24,8,22,52,146,230,188,152,160,164,170,116,32,12,18,30,30,22,0,
0,420,246,410,432,382,318,260,218,192,186,194,182,162,140,152,142,182,168,108,56,34,48,62,108,234,120,126,124,86,60,122,152,148,44,96,116,88,6,62,60,104,74,108,140,20,84,70,68,76,100,102,140,174,86,18,190,488,388,96,80,62,144,124,512,646
,596,568,354,564,644,204,408,598,416,140,32,42,26,24,12,64,154,162,128,132,132,122,70,32,18,14,20,8,12,0,
0,434,242,338,416,420,398,370,358,290,242,250,284,216,224,292,304,212,140,62,20,8,10,32,156,244,100,104,78,62,44,144,92,102,268,316,276,224,180,184,134,42,40,58,152,72,94,138,126,122,118,114,142,254,210,44,140,430,504,322,150,70,148,252,
66,210,508,718,710,494,234,242,440,232,110,72,56,70,60,46,36,16,52,94,76,74,62,42,60,56,42,26,30,6,14,0,
0,394,198,218,280,292,278,268,310,334,266,288,264,238,236,358,488,426,188,86,86,82,74,64,156,246,84,114,92,60,22,78,140,260,158,24,88,156,144,136,244,242,94,46,106,200,204,188,174,174,172,170,176,306,378,232,10,234,398,406,340,258,190,24
2,330,370,382,124,432,246,358,462,178,316,330,164,100,120,116,104,102,90,48,2,16,10,20,46,72,78,70,52,32,18,18,0,
0,332,136,122,164,180,162,136,160,206,206,216,204,188,186,326,528,624,382,164,146,144,138,154,234,274,70,140,158,70,88,88,258,118,76,136,130,176,366,296,36,118,164,78,218,382,280,186,178,182,182,180,184,264,394,428,294,96,104,246,306,372
,366,316,376,512,718,544,696,734,500,370,338,100,250,282,186,154,158,156,166,156,122,74,48,62,72,62,74,84,88,74,52,36,32,0,0,276,78,46,70,88,72,30,30,62,102,130,122,126,122,228,416,626,522,244,154,156,152,170,210,180,28,174,222,112,114,1
64,172,80,156,150,100,72,174,340,140,62,118,170,398,430,224,158,158,164,162,156,172,206,270,366,414,376,262,120,10,128,258,282,302,430,682,662,804,654,246,204,308,358,224,14,130,140,148,154,180,188,160,112,74,86,86,64,70,82,98,92,72,60,4
4,0,
0,238,48,46,70,108,124,90,90,56,64,54,40,60,86,92,266,504,572,338,140,120,128,102,132,120,90,262,186,112,62,148,104,32,88,128,138,126,172,172,158,126,114,356,600,358,156,124,126,124,116,114,134,172,198,220,262,318,352,360,352,302,208,112
,40,102,212,272,480,156,94,74,186,202,104,106,106,116,130,140,162,184,168,120,76,80,74,54,64,78,98,100,90,84,58,0,
0,202,38,70,108,158,188,150,184,170,88,34,28,64,86,70,150,358,532,410,136,60,98,142,172,76,268,298,52,142,38,128,98,78,40,42,140,34,126,126,46,100,188,532,594,214,122,90,86,78,76,76,94,124,142,156,148,134,144,194,262,296,308,294,236,264,
108,228,54,282,220,192,116,214,208,272,174,94,118,136,146,172,174,128,76,64,56,48,56,72,92,94,88,86,84,0,
0,164,30,102,142,182,210,150,202,220,132,62,72,108,142,90,58,240,464,488,206,30,136,192,144,266,366,182,46,88,20,126,86,42,28,32,90,136,178,154,30,70,332,600,384,98,76,34,38,30,32,38,60,84,82,88,96,82,46,34,70,90,108,138,114,138,204,292,
108,330,154,224,132,150,386,202,110,84,102,126,128,150,172,132,70,42,36,36,46,66,90,96,92,98,94,0,
0,140,42,130,162,194,216,116,176,196,114,40,74,114,168,122,56,148,358,516,330,70,184,22,342,374,242,46,68,56,58,132,202,100,82,72,44,220,82,234,120,138,468,486,122,78,58,30,42,38,38,40,64,90,82,74,64,44,22,18,54,68,58,42,6,36,142,364,146
,228,136,82,54,104,140,368,150,44,70,100,100,108,144,120,54,24,18,18,20,52,82,100,108,110,92,0,
0,114,70,154,176,202,206,74,178,192,120,60,60,104,160,138,62,58,218,424,388,90,84,288,338,190,120,90,114,62,104,94,218,372,252,64,62,56,174,62,220,252,502,324,110,112,90,64,78,76,72,74,100,124,114,96,72,48,22,26,64,72,58,36,6,34,24,204,2
76,62,208,42,142,192,200,304,182,10,28,66,68,68,98,94,42,10,6,6,10,34,66,88,108,120,96,0,
0,102,108,178,188,202,176,52,186,208,148,80,54,124,168,162,116,92,192,298,328,68,152,322,72,150,178,152,52,10,92,74,46,194,440,252,126,52,42,194,56,194,402,154,96,100,82,68,84,84,82,86,106,124,110,80,50,26,6,26,58,66,60,48,38,56,46,62,17
6,264,138,102,278,214,296,190,104,50,52,92,88,82,98,82,50,18,12,16,18,38,68,90,116,140,128,0,
0,92,150,200,196,188,126,38,148,184,144,84,98,172,210,200,170,154,248,332,230,134,232,90,74,76,66,52,90,46,128,78,34,74,150,414,152,118,42,36,102,242,298,72,76,80,64,64,76,80,82,80,84,90,72,44,26,12,8,22,56,70,70,64,70,78,68,52,64,200,13
2,138,138,284,306,62,48,50,54,82,70,64,78,86,68,42,34,38,42,60,86,110,140,168,168,0,
0,60,100,214,194,152,58,44,80,128,126,100,126,200,252,252,240,294,352,346,238,226,152,52,166,70,20,118,144,60,182,100,98,10,48,138,216,24,120,110,228,350,264,48,42,50,40,50,52,56,58,48,40,40,30,30,20,4,4,16,44,66,72,70,88,88,76,60,64,110
,246,80,162,112,136,34,20,28,36,54,50,58,78,90,80,60,54,56,60,78,100,122,152,180,184,0,
0,120,228,218,168,86,20,82,84,48,52,48,74,150,236,282,378,386,308,186,138,172,98,122,92,68,28,170,108,44,210,162,84,120,90,90,128,78,92,230,180,338,226,56,38,14,14,26,22,20,30,24,16,28,42,52,56,48,42,46,60,72,74,84,90,84,70,66,68,72,194,
102,170,122,46,20,4,14,22,32,40,58,74,86,80,66,64,68,70,84,104,122,144,168,174,0,
0,182,252,188,94,18,80,98,84,60,36,32,40,102,208,336,366,282,250,104,94,156,120,72,80,86,120,172,68,34,198,262,112,62,58,194,304,136,44,112,368,374,136,58,42,26,26,30,34,40,52,56,52,56,58,64,74,76,74,74,78,78,74,78,76,68,64,66,66,68,98,1
66,172,132,34,14,8,16,12,14,42,56,68,76,74,70,70,74,76,86,102,116,130,146,154,0,
0,250,276,136,66,68,100,86,62,36,24,44,84,106,176,240,190,196,258,168,150,202,148,4,22,200,290,152,28,48,128,228,146,168,196,118,244,218,82,346,576,344,70,56,50,48,52,54,60,62,64,64,62,66,66,64,68,74,72,68,72,72,68,66,62,54,52,54,58,62,6
2,60,52,44,26,10,10,16,10,12,28,42,56,64,64,64,64,66,68,78,92,104,116,126,134,0,
0,340,324,106,130,100,94,60,32,8,4,14,70,154,232,248,146,172,380,106,268,224,60,46,214,418,226,90,14,44,68,166,74,100,112,286,192,238,102,350,364,162,28,32,34,34,36,38,40,40,44,46,46,48,48,46,46,52,52,48,52,52,50,52,46,42,40,38,42,46,48,
50,46,38,32,28,28,32,32,28,32,40,50,54,52,54,56,58,58,62,72,82,94,102,108,0,
0,416,298,154,152,126,82,34,20,12,12,8,26,112,228,294,250,250,388,20,258,140,110,92,368,200,42,50,56,38,38,144,110,124,196,132,148,190,44,370,392,102,4,6,6,8,10,8,10,8,10,12,14,18,18,16,18,24,24,20,22,20,20,26,24,24,26,24,26,26,26,30,30,
28,30,32,30,28,36,32,34,36,38,40,38,40,44,46,48,48,50,56,64,70,72,0,
0,350,108,190,210,140,70,20,10,10,10,8,2,42,144,234,232,220,286,58,220,120,110,324,336,36,8,32,36,18,30,100,158,106,262,106,98,102,162,516,460,54,8,10,8,10,8,6,6,4,4,4,4,8,8,6,8,8,4,2,4,2,2,8,2,4,6,4,8,10,8,10,10,8,10,12,14,12,10,14,16,1
6,16,18,18,18,20,22,26,28,28,30,30,32,34,0,
0,346,174,260,208,118,48,8,6,8,10,8,4,6,60,136,162,156,144,170,194,54,292,426,152,20,34,48,42,4,8,94,188,96,232,198,22,10,158,434,278,22,14,12,10,14,12,10,10,10,12,12,14,12,12,14,14,14,12,8,6,12,18,12,10,10,8,8,10,10,10,10,10,6,8,8
,4,4,6,6,4,4,4,4,6,6,8,8,8,8,4,4,6,0,
0,350,214,226,178,80,18,10,2,10,20,18,10,8,8,48,76,68,10,218,160,144,378,202,92,104,118,80,32,8,46,106,162,122,206,398,94,100,42,550,464,98,10,8,8,12,12,14,12,12,14,12,14,16,14,16,16,14,14,12,14,12,18,18,10,12,10,8,8,12,14,14,14,12,10,8,
12,12,8,8,12,12,8,8,10,14,12,12,14,12,14,14,10,12,12,0,
```

```
28,30,32,30,28,30,32,34,36,38,40,38,40,44,46,48,48,50,56,64,70,72,0,
0,350,108,190,210,140,70,20,10,10,10,8,2,42,144,234,232,220,286,58,220,120,110,324,336,36,8,32,36,18,30,100,158,106,262,106,98,102,162,516,460,54,8,10,8,10,8,6,6,4,4,4,4,8,8,6,8,8,4,2,4,2,2,8,2,4,6,4,8,10,8,10,10,8,10,12,14,12,10,14,16,1
6,16,18,18,18,20,22,26,28,28,30,30,32,34,0,
0,346,174,260,208,118,48,8,6,8,10,8,4,6,60,136,162,156,144,170,194,54,292,426,152,20,34,48,42,4,8,94,188,96,232,198,22,10,158,434,278,22,14,12,10,14,12,10,10,10,12,12,14,12,12,14,14,14,12,8,6,12,18,12,10,10,8,8,10,10,10,10,10,6,8,8
,4,4,6,6,4,4,4,4,6,6,8,8,8,8,4,4,6,0,
0,350,214,226,178,80,18,10,2,10,20,18,10,8,8,48,76,68,10,218,160,144,378,202,92,104,118,80,32,8,46,106,162,122,206,398,94,100,42,550,464,98,10,8,8,12,12,14,12,12,14,12,14,16,14,16,16,14,14,12,14,12,18,18,10,12,10,8,8,12,14,14,14,12,10,8,
12,12,8,8,12,12,8,8,10,14,12,12,14,12,14,14,10,12,12,0,
0,152,72,62,34,84,190,290,194,262,90,14,10,4,6,6,0,8,56,212,334,334,144,30,26,32,50,42,54,42,4,50,80,150,190,158,412,172,92,192,336,214,6,4,2,4,6,6,4,8,12,14,12,10,8,10,8,4,4,4,0,4,8,6,6,6,4,4,6,10,12,10,10,10,6,6,10,10,8,10,14,12,12,14,
14,16,18,22,20,18,20,14,12,16,14,0,
0,0,0,
```

Resultado del Sobel - ESP32

Resultado Sobel - ESP32